Riya Basak

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Research Interest

I investigate **self-supervised world models for embodied intelligence**, focusing on reliable planning and transferable physical knowledge under distribution shift. My long-term goal is **human- and animal-like general intelligence**: understanding how inductive priors and experience-driven learning support causal reasoning, structured memory, and compositional skill transfer across unfamiliar environments.

Education

Sep. 2023 – May 2026 **University of Hertfordshire** Hatfield, United Kingdom

B.Sc. (Hons) Computer Science with Artificial Intelligence — **First-Class Honours**

- **GPA:** 4.38/4.50 cumulative; 4.44/4.50 at Level 6.
- **Capstone Project 1:** *Mitigating Shortcut Learning in Brain Tumour MRI Classification*;

Advisor: Dr Kheng Lee Koay.

- **Capstone Project 2:** *LAADAN-AC: Admissibility-Aware Offline Reinforcement Learning for ICU-Sepsis*

2026 **Highest in Dept. for Final-Year Project** (93% report; 100% demonstration/viva).

2026 **Highest in Course** (LAADAN-AC coursework: 91%).

Research Experience

Jul. 2026 – Present **R&D, Open-source Hugging Face**

- **[Research focus] Transferable physical world models:** Investigating why predictive world models and embodied policies fail under changes in scene geometry, task composition, physical interactions, robot embodiment and dataset distribution.
  - *[Ongoing development]* Developing candidate structured latent dynamics and mechanism-transfer evaluations to distinguish appearance shifts from changes in physical interactions, targeting long-horizon generalisation across unseen environments.
  - *[Research engineering]* Working with LeRobot's synchronised video/state/action data, policy interfaces and simulator evaluation; checking episode integrity and studying the grounding of 2-D observations in 3-D embodied control.

May 2026 – Oct. 2026 **Intelligent Adaptive Systems Research, University of Hertfordshire**

- **[Research Question] What makes an offline treatment policy reliable?** Studied **return, benchmark-defined action admissibility and expert-policy alignment jointly**, because high return alone is insufficient.

Advisor: Dr Manal Helal.

  - *[Contribution]* Extended **LAADAN-AC** with a masked actor, twin reward critics, conservative and expert-policy regularisation, an action-smoothness proxy and Lagrangian cost control; evaluated training-time admissibility against post-hoc filtering.
  - *[Results]* Five fixed final-epoch seeds (95% CIs): **0.7927** $\pm$ **0.0012** return, **0%** mask violations and **0.9540** $\pm$ **0.0032** expert agreement. Compared with post-hoc masked VOAC: **+0.0275** $\pm$ **0.0021** return and **+0.2568** $\pm$ **0.0022** expert agreement.
  - *[Finding]* Ablations identified masking as the direct benchmark-feasibility mechanism; other terms changed the return–alignment trade-off. Tested exploratory portability on a constructed eICU-CRD Demo MDP.

Honors & Awards

2026 **University Graduation Prize**, University of Hertfordshire — highest Level 6 achievement in B.Sc. Computer Science (Artificial Intelligence).

Publications

Conference Paper

C1 **Riya Basak** and Manal Helal. "LAADAN-AC: Beyond Survival in Admissible Offline Treatment-Policy Learning." *International Conference on Advanced Technology and Applied Sciences (ICaTAS)* (2026). **Accepted and orally presented**. Paper website.

Preprint

P1 **Riya Basak**. "Mitigating Shortcut Learning in Brain Tumour MRI Classification." *preprint* (2026). Paper website.

Ongoing Research

Oct. 2026 – Present
When Better Prediction Hurts Planning — Target: TMLR

- Investigating and aiming to address unreliable decisions during test-time world-model adaptation, building on AdaJEPA/OpTTA and related SOTA. Evaluating harmful updates and planning failures under visual, dynamics, and recurring distribution shifts in 2-D benchmarks, with planned extension to 3-D MuJoCo/RoboCasa365 household manipulation and multi-stage tasks to study embodied generalisation.

Nov. 2026 – Present
The World Remembers Contact — TMLR

- Investigating and aiming to address failures in transferring physical dependencies across compositional robot tasks: studying how locally successful interactions alter hidden downstream constraints, beyond existing contact-aware planning, failure diagnosis, and memory, under unfamiliar dynamics and skill combinations.

Professional Experience

Sep. 2025 – Nov. 2025
Mercor, Artificial Intelligence Researcher — Mathematical Reasoning — San Francisco, CA, USA (Remote)

- **[Research Exp.] Generalisation in Olympiad-level mathematics:** Investigated model reasoning on IMO-style and unfamiliar multi-step problems, assessing whether solutions follow valid deductions rather than memorised patterns.
  - *[Contribution]* Designed and audited problem-solving traces and generated proofs through explicit assumptions, constructive arguments, case splits, counterexamples and completeness checks.
  - *[Research outcome]* Identified unsupported implications, missing cases and circular proofs; formalised recurring failures into criteria for evaluating mathematical reasoning and its generalisation.

Aug. 2025 – Sep. 2025
Cubble, Full Stack AI Intern — AI Safety R&D — Edinburgh, Scotland (Remote)

- **[Field / Research Exp.] AI ethics, abuse prevention and chat privacy:** Investigated misuse of AI-assisted features and multimodal content for cyberbullying, harassment, attempted cyberattack facilitation and exposure of private conversations; studied intervention against unsafe requests and repeated abuse.
  - *[Contribution]* Developed policy-driven safety controls for text, image and audio content, linking risk flags to restrictions on unsafe AI use and blocking or escalation of persistent abusive behaviour.
  - *[Field outcome]* Integrated moderation and enforcement into deployed social-chat workflows, enabling traceable interventions and privacy-conscious review of model behaviour and sensitive user content.

Jul. 2025 – Aug. 2025
Cubble, Machine Learning Engineer Intern — Edinburgh, Scotland (Remote)

- **[Research Exp.] Visual–language representation learning for content safety:** Investigated image-only recognition under limited supervision, comparing ResNet50 transfer learning with OpenAI CLIP's text-aligned visual representations for semantic transfer in safe/unsafe content classification.
  - *[Contribution]* Implemented and tuned both classification pipelines; analysed held-out errors, validation stability and overfitting to assess representation quality and generalisation.
  - *[Result]* Achieved **97.5% held-out image accuracy** in the reported evaluation; during integration testing in the application's chat interface, the classifier also identified unseen, slightly blurred and harmful-content images. Recognised by the supervisor as the leading working-model implementation.